SkillNavora

COURSE ADDENDUM

AI Data Engineering: completion addendum

The gap-check topics added to each course module, with the sources used.

Existing lessons stay as they are; this addendum adds to them.

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AI Data Engineering course: completion addendum

This addendum adds the topics the course was missing, mapped to each course module. The existing lessons, handbooks and PDFs are unchanged. Every entry is taken from the Course Gap Check supplements.

Coverage before and after

Prompt module	Course home	Status
1. Advanced Excel for Data Analysis	Course Module 1: Advanced Excel (12 lessons)	COVERED
2. SQL for Data Analysis and Database Management	Course Module 2: SQL (15 lessons)	PARTIAL
3. Power BI and Business Intelligence	Course Module 3: Power BI (6 lessons)	PARTIAL
4. AI and Data Analysis	Course Module 5: AI for Data Analysis (4 lessons)	PARTIAL
5. Python Programming for Data Science	Course Module 6: Python for Data Science (29 lessons)	COVERED
6. Statistics and Probability	Course Module 7: Statistics & ML (first lessons)	COVERED
7. Machine Learning and Predictive Analytics	Course Module 7: Statistics & ML (remaining lessons)	COVERED
8. Data Engineering Fundamentals	NOT IN THE COURSE	MISSING
9. Data Warehousing, ETL and Data Pipelines	NOT IN THE COURSE	MISSING
10. Cloud Computing and Modern Data Platforms	NOT IN THE COURSE	MISSING
11. Generative AI and Large Language Models	Course GenAI track (12 lessons)	COVERED
12. AI Agents and Agentic Workflows	Course Agents track and GenAI lesson 7	PARTIAL
13. MCP, APIs and Integrations for PMs and Business Stakeholders	Course Agents track (APIs, MCP, integrations)	PARTIAL
14. AI and No-Code Tools Laboratory	Course tools track (2 lessons: Figma, n8n)	PARTIAL
15. Data Governance, Security and Responsible AI	Scattered: guardrails (GenAI), fairness and XAI (ML), RLS (Power BI), security (SQL)	MISSING
16. Business Problem-Solving and Analytics Consulting	NOT IN THE COURSE as one module	MISSING
17. Capstone Projects and Portfolio Development	Course capstone lessons (one per track) and Career track (4 lessons)	PARTIAL
18. Interview Preparation and Career Readiness	Career track (4 lessons) and site interview pages	PARTIAL
Part C: 20 cross-functional capabilities	Spread across the course	PARTIAL
Part D: 12-month roadmap	Career lesson 4 (90-day plan) only	PARTIAL
Part E: Tools & Learning Resources directory	Site: Tools & learning resources page	PARTIAL
Part G: PDF 1 Master Handbook	Course PDFs: Complete Data Science & AI Handbook Parts 1 to 4	PARTIAL
Part G: PDFs 2 to 6 (Excel, SQL, Power BI, Python, Stats & ML)	Course lesson PDFs by track	COVERED

Part G: PDF 7 Data Engineering and Cloud Handbook	NOT IN THE COURSE	MISSING
Part G: PDF 8 to 9 (GenAI, RAG, Agents and MCP)	Course GenAI and Agents lesson PDFs	COVERED
Part G: PDF 10 AI and No-Code Tools Laboratory	Tools track (2 lessons)	PARTIAL
Part G: PDF 11 Capstone Projects and Portfolio Workbook	Career lessons and capstone lessons	PARTIAL
Part G: PDF 12 Interview Preparation Handbook	Interview-Handbook.pdf and career interview lessons	COVERED
Part G: PDF 13 12-Month Roadmap and Tracker	NOT IN THE COURSE	MISSING
Part G: PDF 14 Quick Reference and Cheat Sheets	NOT IN THE COURSE	MISSING
Part G: PDF 15 Data Science and AI Glossary	NOT IN THE COURSE	MISSING

Module 1: Advanced Excel

What is added: Excel automation, Flash Fill, Text to Columns and Excel project briefs.

B1 Excel Automation and Missing Excel Skills (course Module 1 gaps)

B1.1

Flash Fill, Text to Columns and standardising data

WHAT IS IT?

> Flash Fill detects a pattern from examples and fills the rest; Text to Columns splits one column into several using a delimiter or fixed width.

WHY IS IT IMPORTANT?

> Analysts spend much of their time cleaning data; these tools remove hours of manual work.

WHERE IS IT USED IN INDUSTRY?

Used in finance, HR and operations reporting to clean exports from systems.

HOW DOES IT WORK?

- Type the result for the first row, then press Ctrl+E (Flash Fill).
- Data > Text to Columns splits by comma, space or fixed width.
- Standardise case with PROPER, UPPER, LOWER and TRIM.

PRACTICAL EXAMPLE

> Full names 'Asha Rao' split into first and last name; phone numbers standardised to +91 format.

TOOL OR SOFTWARE

Microsoft Excel (Windows or Microsoft 365 web)

HANDS-ON EXERCISE

Clean a list of 200 customer names and phone numbers; document each step.

COMMON MISTAKES

Using Flash Fill on data with mixed patterns without checking the result.

INTERVIEW QUESTION

How would you split a column of 'City, State' values in Excel?

OFFICIAL AND FURTHER READING

- Microsoft: Flash Fill (official)

B1.2

Excel macros and VBA: recording and writing automation

WHAT IS IT?

> A macro is a recorded or written set of steps. VBA (Visual Basic for Applications) is the language behind macros in Excel.

WHY IS IT IMPORTANT?

> Repetitive monthly reports can run in seconds instead of hours, and a macro can be checked and reused.

WHERE IS IT USED IN INDUSTRY?

Common in finance, operations and back-office teams.

HOW DOES IT WORK?

- Developer tab > Record Macro, perform the steps, stop recording.
- Alt+F11 opens the VBA editor to read or write code.
- Save macro-enabled files as .xlsm and sign or review macros from unknown sources.

PRACTICAL EXAMPLE

> A macro formats a monthly sales sheet, removes blank rows and saves it as a PDF.

TOOL OR SOFTWARE

Microsoft Excel VBA editor

HANDS-ON EXERCISE

Record a macro that formats a table and then edit its code to add a loop over three sheets.

COMMON MISTAKES

Recording macros without reading the code; hard-coded cell addresses break when data moves.

INTERVIEW QUESTION

What is the risk of running macros from an unknown file?

OFFICIAL AND FURTHER READING

- Microsoft VBA documentation (official)

B1.3

Office Scripts and AI-assisted Excel formulas

WHAT IS IT?

> Office Scripts are TypeScript automations for Excel on the web. AI tools can draft formulas, but each formula must be checked.

WHY IS IT IMPORTANT?

> Office Scripts run in Microsoft 365 without desktop macros, and can be triggered by Power Automate.

WHERE IS IT USED IN INDUSTRY?

Used in Microsoft 365 teams to automate reports that refresh on a schedule.

HOW DOES IT WORK?

- Automate > New Script in Excel on the web.
- Scripts can be run by Power Automate on a schedule.
- AI-written formulas are tested on a sample before use.

PRACTICAL EXAMPLE

> A script creates a summary sheet each Monday; Power Automate runs it and emails the file.

TOOL OR SOFTWARE

Excel on the web, Power Automate

HANDS-ON EXERCISE

Write an Office Script that adds a totals row and run it.

COMMON MISTAKES

Trusting an AI formula without checking edge cases such as blanks and errors.

INTERVIEW QUESTION

What is an Office Script and how does it differ from VBA?

OFFICIAL AND FURTHER READING

- Office Scripts documentation (official)

B1.4

Excel business projects from the prompt (not yet in the course)

WHAT IS IT?

> Projects: customer complaint analysis, financial performance tracker, employee productivity dashboard, customer segmentation and sales performance dashboard.

WHY IS IT IMPORTANT?

> Portfolio projects show the skills to employers, with real business questions.

WHERE IS IT USED IN INDUSTRY?

Analyst roles in every industry.

HOW DOES IT WORK?

- Brief: business question, data, deliverable, metric.
- Clean, model, analyse, visualise, recommend.
- Write a one-page summary for a manager.

PRACTICAL EXAMPLE

> Complaint analysis: count complaints by category and month, flag repeat customers, recommend two fixes with expected impact.

TOOL OR SOFTWARE

Excel

HANDS-ON EXERCISE

Complete the complaint analysis brief with a sample dataset you create.

COMMON MISTAKES

Dashboards with no question or recommendation.

INTERVIEW QUESTION

How would you analyse customer complaints for a telecom company?

OFFICIAL AND FURTHER READING

- Microsoft Excel training (official)

Module 2: SQL

What is added: Recursive CTEs, percentiles, cohort and churn queries, incremental loads and database platforms.

B2 SQL Extras for Analysts and Data Engineers (course Module 2 gaps)

B2.1

Recursive CTEs and hierarchies

WHAT IS IT?

> A recursive CTE repeats a query until no new rows appear, used for hierarchies such as organisation charts or bill-of-materials.

WHY IS IT IMPORTANT?

> Hierarchical data appears in HR, product and supply chain systems.

WHERE IS IT USED IN INDUSTRY?

Org charts, category trees and supply chains.

HOW DOES IT WORK?

- WITH RECURSIVE tree AS (anchor UNION ALL recursive step).
- The recursive step joins the CTE to the table again.
- Always include a stop condition to avoid endless loops.

PRACTICAL EXAMPLE

> WITH RECURSIVE reports AS (SELECT id, manager_id, 1 AS level FROM staff WHERE manager_id IS NULL UNION ALL SELECT s.id, s.manager_id, r.level+1 FROM staff s JOIN reports r ON s.manager_id = r.id) SELECT * FROM reports;

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Build the reporting-level query for a staff table of 20 people.

COMMON MISTAKES

Forgetting the stop condition.

INTERVIEW QUESTION

How would you find everyone who reports to a manager at any depth?

OFFICIAL AND FURTHER READING

- PostgreSQL WITH queries (official)

B2.2

Percentiles, medians and distribution queries

WHAT IS IT?

> Percentiles show the value below which a given share of rows fall. PERCENTILE_CONT interpolates; PERCENTILE_DISC picks an actual value.

WHY IS IT IMPORTANT?

> Medians and percentiles describe skewed business data (order values, delivery times) better than averages.

WHERE IS IT USED IN INDUSTRY?

Used for SLA reporting and pricing analysis.

HOW DOES IT WORK?

- PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY amount) is the median in PostgreSQL.
- Check the database's exact syntax; platforms differ.

PRACTICAL EXAMPLE

> The 90th percentile of delivery time per city, to report SLA breaches.

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Compute median and 90th percentile order value per region on a sample of 10,000 orders.

COMMON MISTAKES

Using AVG for skewed data without checking the median.

INTERVIEW QUESTION

Why report the median rather than the mean for delivery times?

OFFICIAL AND FURTHER READING

- PostgreSQL aggregate functions (official)

B2.3

Business SQL: cohort, funnel, churn, CLV and retention

WHAT IS IT?

> Cohort analysis groups customers by first purchase month and tracks their activity over time. Funnel analysis counts users at each step. Churn is the share lost in a period. CLV (customer lifetime value) estimates profit per customer over time.

WHY IS IT IMPORTANT?

> These are the core metrics in e-commerce, SaaS, banking and telecom analytics.

WHERE IS IT USED IN INDUSTRY?

Used by growth, marketing and customer success teams.

HOW DOES IT WORK?

- Cohort: first_month per customer, then count active customers by months since first purchase.
- Funnel: count distinct users per step, then divide.
- Churn: customers active last period and not this period, divided by active last period.
- CLV (simple): average order value x purchase frequency x expected years.

PRACTICAL EXAMPLE

> Cohort retention table: rows are first-purchase months, columns are months since first purchase, values are the share still buying.

TOOL OR SOFTWARE

PostgreSQL, sample e-commerce data

HANDS-ON EXERCISE

Build a cohort table for 12 months of sample orders and explain the drop after month 1.

COMMON MISTAKES

Counting orders instead of distinct customers.

INTERVIEW QUESTION

How do you calculate monthly churn with SQL?

OFFICIAL AND FURTHER READING

- Mode SQL tutorials: cohort analysis (official learning site)

B2.4

Incremental loading and data-quality SQL

WHAT IS IT?

> Incremental loading processes only new or changed rows using a watermark. Data-quality SQL tests check duplicates, nulls and orphans.

WHY IS IT IMPORTANT?

> Full reloads become too slow and expensive as data grows.

WHERE IS IT USED IN INDUSTRY?

Standard in data warehouse loading.

HOW DOES IT WORK?

- Store the last loaded timestamp.
- Select rows with updated_at greater than the watermark.
- Check: SELECT key, COUNT(*) FROM t GROUP BY key HAVING COUNT(*) > 1.

PRACTICAL EXAMPLE

> Load only orders updated since yesterday, then run a duplicate check on order_id.

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Build an incremental load with a watermark table and two quality checks.

COMMON MISTAKES

Using a watermark based on load time instead of source update time.

INTERVIEW QUESTION

How do you load only new data into a warehouse table?

OFFICIAL AND FURTHER READING

- PostgreSQL SQL reference (official)

B2.5

Database platforms: SQL Server, BigQuery, Snowflake and Databricks SQL

WHAT IS IT?

> The same SQL idea works on several platforms; each has its own functions, pricing and performance features.

WHY IS IT IMPORTANT?

> Job descriptions name platforms; you should know the differences in syntax, pricing model and tuning.

WHERE IS IT USED IN INDUSTRY?

Enterprises use SQL Server; cloud analytics teams use BigQuery, Snowflake or Databricks SQL.

HOW DOES IT WORK?

- SQL Server: T-SQL, SSMS, indexes and columnstore.
- BigQuery: pay per bytes scanned; partition and cluster tables.
- Snowflake: virtual warehouses, pay per compute time.
- Databricks SQL: SQL over Delta tables.

PRACTICAL EXAMPLE

> The same daily revenue query runs on BigQuery with partitioned tables to cut scan cost.

TOOL OR SOFTWARE

BigQuery sandbox (check current terms), Snowflake trial (check terms)

HANDS-ON EXERCISE

Run one query on two platforms and compare how they are priced and tuned.

COMMON MISTAKES

Writing SELECT * on pay-per-scan platforms.

INTERVIEW QUESTION

How does BigQuery pricing change how you write queries?

OFFICIAL AND FURTHER READING

- BigQuery documentation (official)
- Snowflake documentation (official)

Module 3: Power BI

What is added: Microsoft Fabric, lakehouses and semantic models.

A10 Microsoft Fabric and Databricks (Modern Data Platforms)

A10.1

Microsoft Fabric: lakehouse, warehouse, dataflows, semantic models

WHAT IS IT?

> Microsoft Fabric is an all-in-one analytics platform: OneLake storage, lakehouses and warehouses, pipelines, dataflows, notebooks and Power BI semantic models in one place.

WHY IS IT IMPORTANT?

> Many Power BI users now build their models on Fabric; knowing its parts helps with Power BI roles.

WHERE IS IT USED IN INDUSTRY?

Used by Microsoft-centric enterprises and Power BI consultancies.

HOW DOES IT WORK?

- OneLake: one logical data lake for the organisation.
- Lakehouse: Delta tables with SQL and Spark access.
- Warehouse: SQL warehouse.
- Semantic model: the Power BI dataset used by reports.
- Deployment pipelines move content from dev to test to prod.

PRACTICAL EXAMPLE

> A sales lakehouse feeds a semantic model; the Power BI report connects in Direct Lake mode.

TOOL OR SOFTWARE

Microsoft Fabric trial (free trial, check current terms)

HANDS-ON EXERCISE

Create a Fabric trial lakehouse, load a CSV and build a semantic model.

COMMON MISTAKES

Assuming trial limits apply in production; check licences and capacity.

INTERVIEW QUESTION

What is a semantic model in Power BI?

OFFICIAL AND FURTHER READING

- Microsoft Fabric documentation (official)

A10.2

Databricks: notebooks, Delta Lake, Spark jobs and Unity Catalog

WHAT IS IT?

> Databricks is a managed Spark platform with notebooks, Delta Lake tables (ACID transactions on files), jobs and Unity Catalog for governance.

WHY IS IT IMPORTANT?

> Large AI and data teams standardise on Databricks.

WHERE IS IT USED IN INDUSTRY?

Common in finance, healthcare and AI product companies.

HOW DOES IT WORK?

- Notebooks run PySpark and SQL.
- Delta tables support MERGE, time travel and schema evolution.
- Unity Catalog controls access and lineage.

PRACTICAL EXAMPLE

> A MERGE into a Delta silver table keeps the table current from a daily file drop.

TOOL OR SOFTWARE

Databricks Free Edition (check current availability)

HANDS-ON EXERCISE

Create a Delta table from a CSV and run a MERGE.

COMMON MISTAKES

Skipping access controls on shared catalogs.

INTERVIEW QUESTION

Why is Delta Lake useful for pipelines that need reruns?

OFFICIAL AND FURTHER READING

- Databricks documentation (official)

Module 4: AI for Data Analysis

What is added: AI analysis checklist and model choice (see the Tokens and Pricing Handbook for costs).

C6 Context Engineering, Model Choice and AI Analytics Checklist

Item	Answer
Context engineering	Choose what goes into each model call: instructions, relevant retrieved passages, short history and tool results. Keep it small and relevant, label each part, and log what was sent.
Choosing and comparing models	Compare OpenAI, Anthropic Claude and Google Gemini on your own test set for quality, latency, context length, data terms and cost. Pricing and model names change, so check the official pricing page on the day you decide.

AI analytics checklist	Define the question. Check the data quality. Ask the AI for a method, then verify the numbers yourself on a sample. Document the prompt and the output. Note limitations and personal data handling before sharing.

Module 5: Python for Data Science

What is added: Automation, logging, type hints and packaging.

B6 Production Python for Automation (course Module 6 gaps)

B6.1

Automation: email, scheduled scripts and report generation

WHAT IS IT?

> Automation scripts run on a schedule, read files or APIs, produce reports and send them to people.

WHY IS IT IMPORTANT?

> Manual reports waste time and are error-prone.

WHERE IS IT USED IN INDUSTRY?

Common in every analytics team.

HOW DOES IT WORK?

- Use a scheduler (cron, Task Scheduler, Airflow).
- Send email with smtplib or a provider API; never store passwords in code.
- Log each run and alert on failure.

PRACTICAL EXAMPLE

> A script runs at 07:00, builds the daily sales PDF and emails it to managers.

TOOL OR SOFTWARE

Python, smtplib, cron

HANDS-ON EXERCISE

Automate a daily CSV summary and log each run.

COMMON MISTAKES

Hard-coded passwords.

INTERVIEW QUESTION

How would you schedule a Python report and handle failures?

OFFICIAL AND FURTHER READING

- Python smtplib (official)
- Python logging (official)

B6.2

Logging, type hints, packaging and code organisation

WHAT IS IT?

> Logging records what a program does. Type hints document expected types and let tools catch mistakes. Packaging makes code installable.

WHY IS IT IMPORTANT?

> Production code must be readable, observable and reusable.

WHERE IS IT USED IN INDUSTRY?

Expected in data engineering and software roles.

HOW DOES IT WORK?

- logging.basicConfig with levels (INFO, WARNING, ERROR).
- def total(values: list[float]) → float.
- pyproject.toml defines the package.

PRACTICAL EXAMPLE

> A pipeline module logs row counts and fails with a clear message on bad input.

TOOL OR SOFTWARE

Python logging, mypy (check docs)

HANDS-ON EXERCISE

Add logging and type hints to one of your scripts and run mypy.

COMMON MISTAKES

Logging secrets.

INTERVIEW QUESTION

What should a production log include?

OFFICIAL AND FURTHER READING

- Python logging HOWTO (official)

Module 6: Statistics and Machine Learning

What is added: Recommendation systems and regression metrics (MAE, RMSE, R-squared).

B7 Recommendation Systems (course Module 7 gap)

B7.1

Recommendation systems: collaborative and content-based

WHAT IS IT?

> Recommenders suggest items. Collaborative filtering uses patterns across users; content-based filtering uses item features; hybrid systems combine both.

WHY IS IT IMPORTANT?

> Recommendations drive sales and engagement in e-commerce, streaming and banking.

WHERE IS IT USED IN INDUSTRY?

Retail, media and fintech product teams.

HOW DOES IT WORK?

- Build a user-item matrix.
- Measure similarity between users or items.
- Evaluate with precision at k and hit rate on held-out interactions.

PRACTICAL EXAMPLE

> Users who bought the same two products get a suggestion of a third product bought by most of them.

TOOL OR SOFTWARE

scikit-learn, pandas, Surprise library (open source)

HANDS-ON EXERCISE

Build an item-based recommender on a sample ratings file and report precision at 5.

COMMON MISTAKES

Recommending items the user already owns.

INTERVIEW QUESTION

How would you recommend products to a new user with no history?

OFFICIAL AND FURTHER READING

- scikit-learn user guide (official)
- Surprise library docs (official)

B7.2

Regression metrics reference: MAE, MSE, RMSE and R-squared

WHAT IS IT?

> MAE is the average absolute error; MSE the average squared error; RMSE its square root, in the same units as the target; R-squared the share of variation explained.

WHY IS IT IMPORTANT?

> Choosing the right error metric matches the business cost of mistakes.

WHERE IS IT USED IN INDUSTRY?

Forecasting and pricing teams.

HOW DOES IT WORK?

- MAE is robust to outliers; RMSE punishes large errors.
- R-squared can mislead when models are very flexible.

PRACTICAL EXAMPLE

> A delivery-time model has RMSE 2.5 days: typical error is about two and a half days.

TOOL OR SOFTWARE

scikit-learn metrics

HANDS-ON EXERCISE

Compute MAE, RMSE and R-squared for a regression and explain each to a manager.

COMMON MISTAKES

Using R-squared alone to judge a model.

INTERVIEW QUESTION

When would you prefer MAE over RMSE?

OFFICIAL AND FURTHER READING

- scikit-learn regression metrics (official)

Module 7: Agents, MCP and APIs

What is added: OpenAPI, OAuth 2.0, MCP security, integration examples and agent evaluation.

B8 APIs, OpenAPI, OAuth and Integration Depth (Module 13 gaps)

B8.1

OpenAPI and Swagger: describing APIs

WHAT IS IT?

> OpenAPI is a standard file (YAML or JSON) that describes an API's endpoints, parameters and responses. Swagger UI shows it as interactive documentation.

WHY IS IT IMPORTANT?

> Integrators and AI tools read OpenAPI files to call APIs correctly.

WHERE IS IT USED IN INDUSTRY?

Every API-first company publishes OpenAPI specs.

HOW DOES IT WORK?

- Paths and methods.
- Parameters and request bodies.
- Responses and schemas.
- Security schemes.

PRACTICAL EXAMPLE

> An OpenAPI file describes GET /orders?status=open and its response schema.

TOOL OR SOFTWARE

Swagger Editor

HANDS-ON EXERCISE

Write an OpenAPI description for two endpoints of your own app.

COMMON MISTAKES

Specs that are not kept in sync with the code.

INTERVIEW QUESTION

What is OpenAPI used for?

OFFICIAL AND FURTHER READING

- OpenAPI specification (official)

B8.2

OAuth 2.0 in depth and secure tokens

WHAT IS IT?

> OAuth 2.0 lets an app act for a user without the user's password. The app gets an access token with limited scopes; the user approves the access.

WHY IS IT IMPORTANT?

> Integrations with Google, Microsoft, Slack and CRMs use OAuth. Mistakes leak data.

WHERE IS IT USED IN INDUSTRY?

Every SaaS integration.

HOW DOES IT WORK?

- Authorisation code flow with PKCE for apps.
- Scopes limit what the token can do.
- Tokens expire; refresh tokens renew them.
- Never put tokens in URLs or logs.

PRACTICAL EXAMPLE

> A Slack bot asks for chat:write scope only; the token is stored in a secrets manager.

TOOL OR SOFTWARE

Postman OAuth helper

HANDS-ON EXERCISE

Get an OAuth token from a sandbox provider and call one endpoint with it.

COMMON MISTAKES

Using long-lived tokens in code.

INTERVIEW QUESTION

What is the difference between authentication and authorisation in OAuth?

OFFICIAL AND FURTHER READING

- OAuth 2.0 simplified (learning site)

B8.3

MCP security and business integration examples

WHAT IS IT?

> MCP servers expose tools to AI clients. Security needs scoped permissions, user consent for sensitive tools, and logs. Integration examples cover CRM, ERP, Jira, Slack and Notion.

WHY IS IT IMPORTANT?

> A badly scoped MCP tool can change business data.

WHERE IS IT USED IN INDUSTRY?

AI-enabled business tools in every sector.

HOW DOES IT WORK?

- Expose read tools first; add write tools with approval.
- Scope each tool to one system and user role.
- Log every tool call.

PRACTICAL EXAMPLE

> An MCP server lets the assistant read Jira tickets but needs approval to close them.

TOOL OR SOFTWARE

MCP documentation (official)

HANDS-ON EXERCISE

Design an MCP tool list for a support team with read and write tools marked.

COMMON MISTAKES

Giving an AI client admin tokens.

INTERVIEW QUESTION

How would you secure an MCP server that writes to a CRM?

OFFICIAL AND FURTHER READING

- Model Context Protocol (official)

C5 AI Agents: LangGraph, CrewAI, AutoGen and Agent Evaluation

Item	Answer
LangGraph: stateful agents as graphs	Nodes are steps, edges decide the next step, and state is saved between steps. Use it when you need control, retries and human approval. Official docs: docs.langchain.com (LangGraph overview).
CrewAI: role-based agent teams	Agents have roles, goals and tasks; a crew runs them in sequence or with a manager. Good for quick prototypes. Official docs: docs.crewai.com.
AutoGen concepts	Conversational multi-agent framework where agents message each other. Understand the concept even if you use another framework. Check the official project for the current version.
Agent evaluation and monitoring	Measure task completion rate, tool success rate, accuracy, latency and cost per task; keep traces of each run; run a regression test set after every change; test for security (prompt injection and tool misuse).

Module 8: No-code and AI tools lab

What is added: Tool evaluation sheet and tool directory.

C7 Tools and Learning Resources Directory (categories from the prompt)

Item	Answer
How to use this directory	Each entry needs: purpose, beginner explanation, official site, official docs, official tutorial, free or paid status, the course module it supports, learning order and an exercise. Check the official pages before you rely on prices or features; record the date you checked.
Categories	1 Excel and spreadsheets; 2 SQL and databases; 3 Power BI and BI platforms; 4 Python and coding environments; 5 Statistics and mathematics; 6 Machine learning; 7 Data engineering; 8 Cloud platforms; 9 Generative AI; 10 RAG and vector databases; 11 AI agents; 12 MCP and API development; 13 No-code automation; 14 Git and development; 15 Product and project management; 16 Portfolio and interview preparation.
Tool evaluation sheet (for AI and no-code tools)	What it does; why use it; who should use it; main features; free, freemium or paid features and limits; strengths; weaknesses; alternatives; practical example; official documentation; exercise.

Module 9: Capstone, portfolio and interviews

What is added: Eight capstone briefs, interview bank, business scenarios, STAR answers and the readiness check.

C1 Data Engineering and Analytics Interview Questions

Item	Answer
ETL vs ELT	ETL transforms before loading; ELT loads raw data and transforms inside the warehouse. ELT suits cloud warehouses and keeps raw data for rebuilds. Choose ETL when data must be masked before it lands.
What is idempotency?	A pipeline is idempotent if running it twice leaves the same result. Use MERGE on a business key, staging tables and run IDs.
Batch vs streaming?	Batch processes data in scheduled chunks with simpler tooling. Streaming processes events continuously for low latency, at higher complexity. Choose by the freshness the business needs.
Star vs snowflake schema?	A star schema has denormalised dimensions and simple joins, which is fast for BI. A snowflake normalises dimensions, saving space and adding joins.
What is a surrogate key and why use it?	A system-generated key for a dimension row, so history (SCD Type 2) and source key changes do not break facts.
How do you handle late-arriving data?	Reprocess the affected window (backfill), use watermarks that allow a lag, and make loads idempotent.
What does a dbt test check?	Built-in tests check not null, unique, accepted values and relationships; custom tests are SQL queries that should return zero rows.
What is Airflow used for?	Scheduling and monitoring workflows as DAGs with dependencies, retries and alerts. It does not process large data itself.
Spark shuffle and partitions?	A shuffle moves data between machines during joins and groupBy; reduce it with good partitioning, broadcast small tables and avoid skew.
Warehouse vs lakehouse?	A warehouse is optimised for SQL on structured data; a lakehouse adds open table formats on cheap storage for both files and SQL.

How do you test a data pipeline?	Unit tests for transformations, data tests for quality, contract tests for source schemas, and a staging run compared with production totals.
What is CDC?	Change data capture reads inserts, updates and deletes from a database log, so downstream systems stay current without full reloads.

C2 Seven Business Scenario Interviews (with answer frameworks)

Item	Answer
Sales declined by 20%. How would you investigate?	Clarify the period and the scope. Check data quality first (did the data load fully?). Split the decline by region, product, channel, customer segment and price. Compare with last year and with marketing and stock events. Form two or three hypotheses, test each, and recommend actions with expected impact.
Customer churn increased. What would you analyse?	Define churn precisely. Build cohorts and see which cohorts churn. Compare churned and retained customers on usage, support tickets, price changes and product fit. Check onboarding funnel drop-offs. Recommend an experiment, not a guess.
A dashboard is slow. How would you optimise it?	Find the slow visuals and queries. Reduce the data model (remove unused columns, aggregate at the right grain), use summary tables or incremental refresh, simplify DAX or SQL, check relationships and cardinality, and cache where the platform allows.
A data pipeline failed. How would you troubleshoot?	Confirm impact and mitigate (re-run or stop downstream). Read the first error, check upstream freshness and schema changes, reproduce on a sample, fix the root cause, add a test or alert, and write a short review.
An AI assistant gives incorrect answers. How would you investigate?	Log the exact prompt, retrieved context and model version. Check whether the right documents were retrieved. Compare answers with a labelled test set. Separate retrieval failures from generation failures. Add failing cases to the evaluation set and fix the cause.
An enterprise customer wants to automate a manual workflow. How would you design it?	Map the current process and the exceptions. Pick the high-volume, low-risk steps to automate first. Design the flow with systems, data, approvals for risky actions, logging and fallbacks. Pilot with success metrics, then scale.
A stakeholder asks for a dashboard but cannot define requirements. How would you proceed?	Ask about the decision the dashboard supports, who uses it and how often. Sketch two or three mock screens, get feedback, agree the first five metrics and a definition for each, then build a first version in a short cycle.

C3 Behavioural Interviews and STAR Answers

Item	Answer
Tell me about yourself	Two minutes: current role, one relevant project with a result, the skills you want to use in this role, and why this company.
Explain a difficult project	Situation (context and stakes), Task (your role), Action (what you did, with trade-offs), Result (measured outcome), Learning.
Describe a problem you solved	Pick a problem with data or a process change. Show how you found the root cause and what changed afterwards.
Explain a conflict with a stakeholder	Show that you understood their goal, aligned on data and the decision, and kept the relationship.
Describe a failed project	Own the failure, explain what you missed, what it cost, and what you do differently now.
How do you prioritise work?	Value, urgency, effort and risk; agree priorities with the owner; say no to low-value requests with a reason.

How do you explain technical concepts to non-technical stakeholders?	Start with the business question, use one visual, use plain words, check understanding, and give the decision needed.

C4 Eight Capstone Project Briefs and Portfolio Checklist

Item	Answer
Project 1: Business analytics dashboard	Excel cleaning, SQL analysis, Power BI dashboard, KPI reporting and recommendations. Deliverables: dataset, cleaning log, SQL file, PBIX, one-page summary.
Project 2: Customer churn prediction	Python, pandas, statistics, a classification model, evaluation with precision, recall and ROC-AUC, and a business interpretation with a retention plan.
Project 3: End-to-end data engineering pipeline	Extract from an API, load raw files, transform with dbt or SQL into bronze, silver and gold layers, orchestrate with Airflow or a scheduler, and run data-quality checks. Deliverables: repository, architecture diagram, run instructions, quality report.
Project 4: AI-powered data analyst	Upload a CSV or Excel, ask questions in natural language, generate and validate SQL or Python, create charts and a business summary. Validation: compare generated results with a manual check.
Project 5: Enterprise RAG knowledge assistant	Ingest documents, chunk, embed, store in a vector index, retrieve, answer with citations, evaluate faithfulness and recall on a labelled question set.
Project 6: AI agent for business operations	Design the agent, add tool calls to two business APIs, orchestrate steps, add human approval for risky actions, and monitor each run.
Project 7: MCP-powered business assistant	An MCP server with read tools for a database and one business app, scoped permissions, logs, and an AI client that uses it.
Project 8: Industry-specific capstone (choose one)	Real estate lead and customer analytics; education analytics and an AI learning assistant; financial document processing and analytics; customer service intelligence; e-commerce customer intelligence. Each needs: business problem, users, data sources, requirements, architecture, data flow diagram, implementation steps, tests, metrics, limitations, future work, GitHub repository, demo video and documentation.
Portfolio checklist (all projects)	GitHub repository with README, requirements file, tests, sample data (or a public dataset link), screenshots, a 3-minute demo video, a one-page business summary and a limitations section.

C9 Career Readiness Check

Item	Answer
Self-check (answer yes or no)	I can write SQL joins, window functions and cohorts from memory. I can build a dashboard with a star schema and DAX measures. I can clean data and explain every step. I can explain a model's metrics to a non-technical person. I have two published projects with READMEs. I can describe an ETL pipeline and its tests. I can explain RAG and a basic agent. I can explain privacy basics for personal data.
Scoring	Ten or more yes: ready to apply for analyst roles with a portfolio. Seven to nine: complete the missing topics in the matching supplement and build one more project. Fewer than seven: return to the foundation months of the roadmap.

Added: Data Engineering and Cloud

What is added: ETL and ELT, pipelines, warehouses and lakehouses, medallion architecture, Airflow, dbt, Spark, Kafka, data quality, cloud services, Docker, Kubernetes and CI/CD.

A1 Data Engineering Fundamentals

A1.1

What data engineering is and the roles

WHAT IS IT?

> Data engineering builds and runs the systems that move, store and prepare data so analysts, data scientists and AI applications can use it reliably.

WHY IS IT IMPORTANT?

> Without reliable pipelines, dashboards show wrong numbers and models train on stale data. Engineers own that reliability.

WHERE IS IT USED IN INDUSTRY?

Banks, retailers, SaaS companies and AI product teams all employ data engineers to feed their reporting, ML and AI features.

HOW DOES IT WORK?

- Sources (apps, databases, APIs, files) produce raw data.
- Ingestion copies it into a landing zone.
- Transformation cleans and models it.
- Serving gives analysts and models trusted tables.

PRACTICAL EXAMPLE

> A sales app writes orders to a database; a nightly pipeline copies them to a warehouse and builds a daily revenue table that the Power BI dashboard reads.

TOOL OR SOFTWARE

Any SQL client and Python

HANDS-ON EXERCISE

Draw the path of one order from the app to the Power BI dashboard, naming each step and the owner.

COMMON MISTAKES

Treating a data engineer as a 'report builder'. Their work is the reliability and repeatability of the flow.

INTERVIEW QUESTION

What is the difference between a data analyst, a data scientist and a data engineer?

OFFICIAL AND FURTHER READING

- Apache Airflow docs (official)
- dbt docs (official)

A1.2

Data types and file formats: CSV, JSON, XML, Parquet, Avro, ORC

WHAT IS IT?

A file format is the way data is written to disk. Row formats (CSV, JSON, Avro) store records one after another; column formats (Parquet, ORC) store each column together.

WHY IS IT IMPORTANT?

Column formats read faster and smaller for analytics, and compression cuts storage cost. Choosing the wrong format is a common cause of slow, expensive pipelines.

WHERE IS IT USED IN INDUSTRY?

Data lakes and warehouses commonly store Parquet; Kafka pipelines often use Avro with a schema registry.

HOW DOES IT WORK?

- CSV: plain text, no types, easy to read.
- JSON: nested, flexible, larger.
- Parquet and ORC: columnar, typed, compressed, best for analytics.
- Avro: row format with a schema, common for streams.
- Compression (Snappy, gzip, zstd) trades CPU for size.

PRACTICAL EXAMPLE

A 2 GB CSV of transactions becomes about 250 MB as Parquet, and a query summing one column reads only that column.

TOOL OR SOFTWARE

pandas, DuckDB

HANDS-ON EXERCISE

Convert a CSV of 1 million rows to Parquet and compare file size and the time to sum one column.

COMMON MISTAKES

Storing everything as CSV because it is familiar, then wondering why queries are slow.

INTERVIEW QUESTION

Why is Parquet better than CSV for analytics?

OFFICIAL AND FURTHER READING

- Apache Parquet documentation (official)
- pandas IO tools (official)

A1.3

Ingestion: batch, streaming, API, file, database and CDC

WHAT IS IT?

> Ingestion is the step that brings data from sources into your platform. Batch loads chunks on a schedule; streaming sends events as they happen; CDC (change data capture) reads database changes from the transaction log.

WHY IS IT IMPORTANT?

> The ingestion method decides freshness, cost and risk of missed or duplicated records.

WHERE IS IT USED IN INDUSTRY?

Fraud teams use streaming; finance reporting usually uses daily batch; inventory systems often use CDC from the operational database.

HOW DOES IT WORK?

- Batch: scheduled pull of new or changed rows (often by an updated_at column).
- Streaming: events published to a topic and consumed continuously.
- API: paginated pulls with rate limits and tokens.
- CDC: reads inserts, updates and deletes from the database log.

PRACTICAL EXAMPLE

> An API returns orders in pages of 100; the pipeline stores the last updated timestamp and asks only for newer orders next run.

TOOL OR SOFTWARE

Python requests, Airbyte (open source connectors), Debezium for CDC

HANDS-ON EXERCISE

Write a script that pulls a paginated public API and stores only new records using a stored cursor.

COMMON MISTAKES

Pulling the whole table every night when only changes are needed.

INTERVIEW QUESTION

What is the difference between batch and streaming ingestion?

OFFICIAL AND FURTHER READING

- Debezium documentation (official)
- Airbyte docs (official)

A1.4

ETL versus ELT

WHAT IS IT?

> ETL extracts, transforms outside the target, then loads. ELT extracts and loads raw data first, then transforms it inside the warehouse.

WHY IS IT IMPORTANT?

> ELT is common on cloud warehouses that have cheap, fast compute, and keeps raw data for re-processing. ETL still fits when data must be cleaned or masked before it lands.

WHERE IS IT USED IN INDUSTRY?

Most modern analytics teams use ELT with dbt; regulated teams may use ETL to remove personal data before loading.

HOW DOES IT WORK?

- ETL: transform in a separate engine, then load clean tables.
- ELT: load raw tables first (for example in a raw schema), then SQL models build the clean layers.
- Keep the raw data so models can be rebuilt.

PRACTICAL EXAMPLE

> Raw Shopify orders load into a raw schema; a dbt model builds a cleaned orders table and a daily revenue table.

TOOL OR SOFTWARE

dbt, SQL

HANDS-ON EXERCISE

Describe for one business report whether you would use ETL or ELT and why, in five sentences.

COMMON MISTAKES

Transforming in place and losing the raw data needed to fix mistakes later.

INTERVIEW QUESTION

When would you choose ETL over ELT?

OFFICIAL AND FURTHER READING

- dbt: what is dbt (official)

A1.5

Pipeline reliability: idempotency, retries, incremental loads

WHAT IS IT?

> An idempotent pipeline gives the same result when run twice. Incremental loads process only new or changed data. Retries repeat a failed step with a wait between attempts.

WHY IS IT IMPORTANT?

> Pipelines fail and are re-run. Without idempotency, a re-run creates duplicate sales and wrong totals.

WHERE IS IT USED IN INDUSTRY?

Every production pipeline in finance, e-commerce and logistics must survive reruns and partial failures.

HOW DOES IT WORK?

- Use MERGE or upsert keyed on a business key instead of plain INSERT.
- Load into a staging table, validate, then swap or merge.
- Store a run ID and watermark so reruns know where they stopped.
- Retry transient errors (timeouts) with exponential backoff; fail fast on bad data.

PRACTICAL EXAMPLE

> A nightly job merges orders on order_id; running it twice still leaves one row per order.

TOOL OR SOFTWARE

SQL MERGE or INSERT ... ON CONFLICT (PostgreSQL)

HANDS-ON EXERCISE

Write a MERGE that upserts 1,000 orders twice and prove the row count is unchanged.

COMMON MISTAKES

Using INSERT for a daily load, so reruns duplicate rows.

INTERVIEW QUESTION

What does idempotent mean for a data pipeline?

OFFICIAL AND FURTHER READING

- PostgreSQL INSERT ... ON CONFLICT (official)

A1.6

Data modelling: conceptual, logical, physical; star and snowflake schemas

WHAT IS IT?

> Conceptual models name the business entities; logical models add attributes and keys; physical models are the actual tables. A star schema has one fact table of measures and dimension tables describing them.

WHY IS IT IMPORTANT?

> A clear model makes queries simple and consistent, so two teams get the same revenue number.

WHERE IS IT USED IN INDUSTRY?

Almost every BI platform (Power BI, Tableau, Looker) works best on a star schema.

HOW DOES IT WORK?

- Fact table: sales lines with amount, quantity and foreign keys.
- Dimension tables: customer, product, date, store.
- Snowflake schema normalises dimensions further (more joins, less duplication).

PRACTICAL EXAMPLE

> fact_sales(order_line_id, date_key, customer_key, product_key, amount) joined to dim_customer, dim_product, dim_date.

TOOL OR SOFTWARE

SQL, Power BI model view

HANDS-ON EXERCISE

Design a star schema for a coffee-shop chain with sales, products, stores and dates; list the grain of the fact table.

COMMON MISTAKES

Mixing grains in one fact table (order-level and line-level measures together).

INTERVIEW QUESTION

What is the grain of a fact table and why does it matter?

OFFICIAL AND FURTHER READING

- Kimball Group: dimensional modelling techniques

A1.7

Slowly Changing Dimensions (SCD) types 1, 2 and 3

WHAT IS IT?

> A slowly changing dimension is an attribute that changes over time, such as a customer's city. Type 1 overwrites it; Type 2 keeps history with valid-from and valid-to dates; Type 3 keeps one previous value.

WHY IS IT IMPORTANT?

> Reports about past periods must use the value that was true then. Type 2 makes that possible.

WHERE IS IT USED IN INDUSTRY?

Customer, employee and product history in banks, HR systems and retailers depend on SCD Type 2.

HOW DOES IT WORK?

- Type 1: update the row; history is lost.
- Type 2: close the old row (valid_to) and insert a new row with a new surrogate key.
- Type 3: add a column for the previous value.

PRACTICAL EXAMPLE

> A customer moves from Pune to Bengaluru on 1 Jan; the old row ends on 31 Dec and the new row starts on 1 Jan, so last year's sales still show Pune.

TOOL OR SOFTWARE

SQL

HANDS-ON EXERCISE

Write a Type 2 update for one customer change and show the two rows.

COMMON MISTAKES

Using Type 1 for attributes that reports need historically.

INTERVIEW QUESTION

What is the difference between SCD Type 1 and Type 2?

OFFICIAL AND FURTHER READING

- Kimball Group: slowly changing dimensions

A2 Data Warehousing, Lakes and Lakehouses

A2.1

Data warehouse, data mart, data lake and lakehouse

WHAT IS IT?

> A data warehouse stores cleaned, structured data for reporting. A data mart is a warehouse slice for one department. A data lake stores raw files of any type cheaply. A lakehouse adds warehouse features (tables, ACID transactions, SQL) on top of lake storage.

WHY IS IT IMPORTANT?

> Choosing the right storage decides cost, speed and who can use the data.

WHERE IS IT USED IN INDUSTRY?

Retailers and banks typically run a warehouse (Snowflake, BigQuery, Redshift); AI teams often use a lake or lakehouse for unstructured data.

HOW DOES IT WORK?

- Warehouse: schema on write, optimised for SQL.
- Lake: schema on read, cheap object storage (S3, Azure Blob, Google Cloud Storage).
- Lakehouse: table formats (Delta Lake, Apache Iceberg) give transactions on lake files.

PRACTICAL EXAMPLE

> Raw clickstream JSON sits in the lake; cleaned sessions go to the warehouse for BI.

TOOL OR SOFTWARE

BigQuery, Snowflake, Databricks (concepts)

HANDS-ON EXERCISE

Write a one-page comparison of warehouse, lake and lakehouse for a company with 3 TB of logs and a BI team.

COMMON MISTAKES

Putting raw logs in the warehouse without cost controls.

INTERVIEW QUESTION

What is the difference between a data lake and a data warehouse?

OFFICIAL AND FURTHER READING

- Snowflake: what is a data warehouse (official)
- Databricks: lakehouse (official)

A2.2

Medallion architecture: bronze, silver and gold

WHAT IS IT?

> Medallion architecture organises a lakehouse or warehouse in three layers: bronze (raw, as received), silver (cleaned and conformed), gold (business-ready tables and metrics).

WHY IS IT IMPORTANT?

> Each layer has one job, so bugs are found and fixed in one place, and analysts use only trusted gold tables.

WHERE IS IT USED IN INDUSTRY?

Used widely on Databricks, Microsoft Fabric and by dbt-based teams.

HOW DOES IT WORK?

- Bronze: append raw data with load time and source.
- Silver: deduplicate, fix types, apply keys and quality rules.
- Gold: aggregates and star schemas for dashboards and models.

PRACTICAL EXAMPLE

> bronze.orders (raw JSON) → silver.orders (typed, deduplicated) → gold.daily_revenue (used by Power BI).

TOOL OR SOFTWARE

dbt, Databricks, Fabric (concepts)

HANDS-ON EXERCISE

Sketch the three layers for an invoices dataset and name one quality check at each layer.

COMMON MISTAKES

Letting analysts query bronze tables that are not validated.

INTERVIEW QUESTION

Explain the medallion architecture.

OFFICIAL AND FURTHER READING

- Databricks: medallion architecture (official)

A3 Orchestration and Pipelines

A3.1

Apache Airflow: DAGs, tasks, scheduling and retries

WHAT IS IT?

> Airflow runs workflows as DAGs (directed acyclic graphs): tasks with dependencies, scheduled and retried, with logs and a web UI.

WHY IS IT IMPORTANT?

> Pipelines have many steps that must run in order and recover from failure. Orchestrators make that visible and repeatable.

WHERE IS IT USED IN INDUSTRY?

Common in data platforms at banks, media and e-commerce companies.

HOW DOES IT WORK?

- Define a DAG in Python with a schedule.
- Each task is an operator (Python, SQL, Bash, cloud service).
- Dependencies are set with >> (for example extract >> transform >> load).
- Retries and alerts are set per task.

PRACTICAL EXAMPLE

> A DAG runs at 02:00: extract orders, run dbt models, run data-quality tests, and alert the team if a test fails.

TOOL OR SOFTWARE

Apache Airflow

HANDS-ON EXERCISE

Install Airflow locally with Docker and build a three-task DAG that prints dates and fails on purpose once.

COMMON MISTAKES

Putting heavy data processing inside the orchestrator instead of calling a warehouse or Spark.

INTERVIEW QUESTION

What is a DAG and why are retries set per task?

OFFICIAL AND FURTHER READING

- Apache Airflow documentation (official)
- Airflow Docker quickstart (official)

A3.2

dbt: models, tests, documentation and lineage

WHAT IS IT?

dbt (data build tool) lets analysts write SQL SELECT statements as models; dbt builds them in the warehouse in order, runs tests (not null, unique, relationships) and generates documentation and lineage graphs.

WHY IS IT IMPORTANT?

dbt brings software practices (version control, tests, docs) to SQL transformations.

WHERE IS IT USED IN INDUSTRY?

Standard in analytics engineering teams; listed in many data analyst and analytics engineer job descriptions.

HOW DOES IT WORK?

- models/*.sql: each file is a table or view.
- ref('model') sets dependencies.
- schema.yml defines tests and descriptions.
- dbt run builds, dbt test checks, dbt docs generate the site.

PRACTICAL EXAMPLE

stg_orders.sql cleans raw orders; fct_orders.sql joins to customers; a test asserts order_id is unique.

TOOL OR SOFTWARE

dbt Core (open source) with DuckDB or PostgreSQL

HANDS-ON EXERCISE

Build two dbt models over a sample orders table and add three tests.

COMMON MISTAKES

Writing one giant model instead of layers (staging, intermediate, marts).

INTERVIEW QUESTION

What do dbt tests check?

OFFICIAL AND FURTHER READING

- dbt documentation (official)
- dbt tutorial (official)

A3.3

Apache Spark and PySpark (big data processing)

WHAT IS IT?

> Spark processes large datasets in parallel across a cluster. PySpark is its Python API using DataFrames.

WHY IS IT IMPORTANT?

> Single machines run out of memory on large data; Spark splits the work and combines results.

WHERE IS IT USED IN INDUSTRY?

Used for large transformations in telecom, retail and AI feature pipelines.

HOW DOES IT WORK?

- DataFrames are distributed tables; transformations are lazy until an action (write, show).
- Partitioning splits data for parallel work; shuffles (joins, groupBy) move data between machines and are the costly step.

PRACTICAL EXAMPLE

```
from pyspark.sql import functions as F; df.groupBy('city').agg(F.sum('amount')).write.parquet(...)
```

TOOL OR SOFTWARE

PySpark (local mode), Databricks Community Edition

HANDS-ON EXERCISE

Run a groupBy on a 10-million-row file locally in PySpark and explain where the shuffle happens.

COMMON MISTAKES

Calling collect() on big data, which pulls everything to the driver.

INTERVIEW QUESTION

What is a shuffle in Spark and why is it expensive?

OFFICIAL AND FURTHER READING

- Apache Spark SQL guide (official)

A3.4

Streaming with Apache Kafka

WHAT IS IT?

> Kafka is a distributed log: producers write events to topics; consumers read them, at their own pace, from partitions.

WHY IS IT IMPORTANT?

> Real-time features (fraud alerts, stock levels, live dashboards) need events processed as they arrive.

WHERE IS IT USED IN INDUSTRY?

Used by banks, logistics and large e-commerce platforms.

HOW DOES IT WORK?

- Topic: a named stream; partitions give parallelism.
- Producer writes; consumer group reads and tracks offsets.
- Retention keeps events for replay.

PRACTICAL EXAMPLE

> Order events go to topic orders; a consumer updates the live inventory table.

TOOL OR SOFTWARE

Kafka (local Docker), Python kafka client

HANDS-ON EXERCISE

Produce 100 events and consume them in a script; then restart the consumer and show it resumes from its offset.

COMMON MISTAKES

Assuming message order across partitions, or not handling duplicate events.

INTERVIEW QUESTION

What is a Kafka partition used for?

OFFICIAL AND FURTHER READING

- Apache Kafka documentation (official)

A4 Data Quality, Observability and Governance Basics

A4.1

Data quality dimensions and rules

WHAT IS IT?

> Data quality is measured on accuracy, completeness, consistency, validity, uniqueness and timeliness. Rules turn these into checks that run on every load.

WHY IS IT IMPORTANT?

> Bad data makes reports wrong; checks catch it before people see it.

WHERE IS IT USED IN INDUSTRY?

Mandatory in finance, healthcare and regulated reporting.

HOW DOES IT WORK?

- Completeness: no missing mandatory fields.
- Uniqueness: one row per business key.
- Validity: values in range and in allowed lists.
- Timeliness: data arrives on schedule.
- Reconciliation: totals match the source.

PRACTICAL EXAMPLE

> Test: order_amount >= 0; test: customer_id exists in customers; test: daily row count within 20% of last week.

TOOL OR SOFTWARE

SQL checks, dbt tests, Great Expectations (open source)

HANDS-ON EXERCISE

Write ten SQL checks for a sales table and run them against a sample with planted errors.

COMMON MISTAKES

Checking quality only at the end of the pipeline, after bad data has spread.

INTERVIEW QUESTION

Name the six dimensions of data quality and give an example of each.

OFFICIAL AND FURTHER READING

- Great Expectations docs (official)
- dbt tests (official)

A4.2

Data observability and pipeline monitoring

WHAT IS IT?

> Data observability tracks freshness, volume, schema changes and distribution of data, and alerts when something looks wrong.

WHY IS IT IMPORTANT?

> Pipelines can succeed technically and still load empty or wrong data. Observability catches that.

WHERE IS IT USED IN INDUSTRY?

Data platform teams use freshness and volume alerts on critical tables.

HOW DOES IT WORK?

- Freshness: last update time against an agreed SLA.
- Volume: row count compared to history.
- Schema: alerts when a column is added, removed or changes type.
- Alerts go to the owner with a link to the failed run.

PRACTICAL EXAMPLE

> The orders table has no new rows by 08:00; an alert opens an incident for the data owner.

TOOL OR SOFTWARE

Airflow alerts, dbt source freshness

HANDS-ON EXERCISE

Set a freshness check on a sample table and trigger it by stopping the load.

COMMON MISTAKES

Only monitoring whether the job ran, not whether the data is right.

INTERVIEW QUESTION

What is data freshness and how would you monitor it?

OFFICIAL AND FURTHER READING

- dbt source freshness (official)

A4.3

Data catalogues, lineage and classification

WHAT IS IT?

> A data catalogue lists datasets with owners and descriptions. Lineage shows where data came from and where it goes. Classification labels data as public, internal, confidential or restricted.

WHY IS IT IMPORTANT?

> People can find trusted data, and impact of a change is known before it breaks reports.

WHERE IS IT USED IN INDUSTRY?

Enterprises use catalogues such as Microsoft Purview, Collibra and Atlan.

HOW DOES IT WORK?

- Register each table with owner, description and tags.
- Lineage graphs come from dbt or the catalogue.
- Classification drives access and retention rules.

PRACTICAL EXAMPLE

> A table tagged PII restricted to HR analysts; lineage shows it feeds a payroll dashboard.

TOOL OR SOFTWARE

dbt docs (lineage), Microsoft Purview (overview)

HANDS-ON EXERCISE

Document one dbt project with descriptions and a lineage graph screenshot.

COMMON MISTAKES

Having no owner for a dataset, so nobody fixes it.

INTERVIEW QUESTION

What is data lineage and why does it matter?

OFFICIAL AND FURTHER READING

- Microsoft Purview documentation (official)

A9 Cloud Fundamentals for Data

A9.1

Cloud basics: IaaS, PaaS, SaaS, regions and identity

WHAT IS IT?

> Cloud computing rents compute, storage and services over the internet. IaaS gives virtual machines; PaaS gives managed platforms; SaaS gives finished applications. Regions and availability zones decide where data lives. Identity and access management (IAM) decides who can do what.

WHY IS IT IMPORTANT?

> Data platforms run in the cloud; wrong permissions or regions cause security and compliance failures.

WHERE IS IT USED IN INDUSTRY?

Almost every data role uses cloud storage, compute and IAM daily.

HOW DOES IT WORK?

- Choose a region close to users and matching data residency rules.
- Give services the least privilege they need (roles, not shared keys).
- Use managed services to reduce operations work.

PRACTICAL EXAMPLE

> A data pipeline runs as a service role that can read one bucket and write to one warehouse schema, nothing else.

TOOL OR SOFTWARE

AWS, Azure and Google Cloud consoles

HANDS-ON EXERCISE

Draw the IAM roles for a pipeline that reads files from storage and writes to a warehouse.

COMMON MISTAKES

Using a root or owner account for daily work, or storing keys in code.

INTERVIEW QUESTION

What is least privilege and why does it matter for data pipelines?

OFFICIAL AND FURTHER READING

- AWS: what is cloud computing (official)
- Google Cloud documentation (official)

A9.2

Data services on AWS, Azure and Google Cloud (mapping)

WHAT IS IT?

> The three big clouds offer similar data services under different names. Knowing the mapping lets you move between them.

WHY IS IT IMPORTANT?

> Customers use different clouds; FDEs and data engineers must read any of them.

WHERE IS IT USED IN INDUSTRY?

Enterprise job descriptions name one or more of these clouds.

HOW DOES IT WORK?

- Object storage: AWS S3, Azure Blob Storage, Google Cloud Storage.
- Serverless ETL: AWS Glue, Azure Data Factory, Google Cloud Dataflow.
- Warehouse: Amazon Redshift, Azure Synapse, Google BigQuery.
- Streaming: Amazon Kinesis or MSK, Azure Event Hubs, Google Pub/Sub.
- Analytics platform: Databricks on all three, Microsoft Fabric on Azure, Snowflake on all three.

PRACTICAL EXAMPLE

> The same medallion pipeline runs on S3 + Glue + Redshift or on Blob + Data Factory + Synapse.

TOOL OR SOFTWARE

Cloud consoles and CLIs

HANDS-ON EXERCISE

Make a table with the service names for storage, ETL, warehouse, streaming and orchestration on each cloud.

COMMON MISTAKES

Assuming a service works the same in every cloud; limits and pricing differ.

INTERVIEW QUESTION

Which cloud data services would you use for a batch warehouse on Azure?

OFFICIAL AND FURTHER READING

- AWS data analytics services (official)
- Azure Data Factory docs (official)
- Google BigQuery docs (official)

A9.3

Cloud cost optimisation for data platforms

WHAT IS IT?

> Cost optimisation keeps cloud spend proportional to value: right-sized compute, auto-suspend, partitioning, storage tiers and budgets.

WHY IS IT IMPORTANT?

> Warehouses and Spark clusters left running cost thousands of rupees or dollars a month for nothing.

WHERE IS IT USED IN INDUSTRY?

Cloud cost is a daily concern for data platform teams and FDE customers.

HOW DOES IT WORK?

- Set budgets and alerts on day one.
- Auto-suspend idle warehouses and clusters.
- Partition and cluster big tables; avoid SELECT * on huge tables.
- Move cold data to cheaper storage tiers.

PRACTICAL EXAMPLE

> A warehouse set to auto-suspend after 5 minutes cuts a monthly bill by most of the idle time.

TOOL OR SOFTWARE

Cloud cost dashboards

HANDS-ON EXERCISE

Estimate the monthly cost of a pipeline that runs hourly for 2 hours and propose three cost cuts.

COMMON MISTAKES

Leaving dev clusters running overnight.

INTERVIEW QUESTION

How would you reduce a warehouse bill that doubled this month?

OFFICIAL AND FURTHER READING

- AWS cost optimization pillar (official)

A11 Containers, Deployment and Operations

A11.1

Docker fundamentals for data work

WHAT IS IT?

> Docker packages an app and its dependencies into an image that runs the same everywhere. A container is a running image.

WHY IS IT IMPORTANT?

> Pipelines and notebooks must run the same on a laptop, in CI and in the cloud.

WHERE IS IT USED IN INDUSTRY?

Standard in data engineering and AI deployment.

HOW DOES IT WORK?

- Dockerfile lists the build steps.
- docker build creates an image; docker run starts a container.
- docker compose starts several services (database, Airflow, app) together.

PRACTICAL EXAMPLE

> A Dockerfile installs Python and dependencies, copies the pipeline code and runs it on a schedule.

TOOL OR SOFTWARE

Docker Desktop (check current licence terms for your use)

HANDS-ON EXERCISE

Containerise a small Python script that reads a CSV and writes Parquet.

COMMON MISTAKES

Baking secrets into the image.

INTERVIEW QUESTION

What is the difference between an image and a container?

OFFICIAL AND FURTHER READING

- Docker get started (official)

A11.2

Kubernetes basics for data platforms

WHAT IS IT?

> Kubernetes runs containers across a cluster: deployments keep a number of pods running, services route traffic, and jobs run tasks to completion.

WHY IS IT IMPORTANT?

> Large data and AI platforms run on Kubernetes; engineers need to read manifests and debug pods.

WHERE IS IT USED IN INDUSTRY?

Used by large enterprises on EKS, AKS and GKE.

HOW DOES IT WORK?

- Pod: one or more containers.
- Deployment: keeps N replicas.
- Service: stable network address.
- Job and CronJob: run batch work.

PRACTICAL EXAMPLE

> A CronJob runs the nightly pipeline container at 02:00 on the cluster.

TOOL OR SOFTWARE

kubectl and a local cluster (kind)

HANDS-ON EXERCISE

Deploy a CronJob that prints the date and check its logs.

COMMON MISTAKES

Running stateful databases without understanding storage.

INTERVIEW QUESTION

What is a Kubernetes CronJob used for?

OFFICIAL AND FURTHER READING

- Kubernetes concepts (official)

A11.3

CI/CD for data pipelines and dbt projects

WHAT IS IT?

> CI/CD runs tests and deploys changes automatically: on each pull request tests run, and on merge the pipeline deploys to the target environment.

WHY IS IT IMPORTANT?

> Data changes break reports; automated tests catch them before production.

WHERE IS IT USED IN INDUSTRY?

Standard for analytics engineering and data platform teams.

HOW DOES IT WORK?

- GitHub Actions workflow: lint, dbt build on a test schema, dbt test.
- Deploy on merge to main.
- Store secrets in the CI secret store, never in code.

PRACTICAL EXAMPLE

> A pull request runs dbt build against a test schema; merge deploys to production.

TOOL OR SOFTWARE

GitHub Actions

HANDS-ON EXERCISE

Write a workflow that runs dbt test on each pull request.

COMMON MISTAKES

Testing only in production.

INTERVIEW QUESTION

How would you add tests for a dbt project in CI?

OFFICIAL AND FURTHER READING

- GitHub Actions documentation (official)

A12 Data Architecture and System Design Basics

A12.1

Data architecture: sources, ingestion, storage, serving and governance

WHAT IS IT?

> A data architecture diagram shows each source, pipeline, store, consumer and control (security, quality, cost).

WHY IS IT IMPORTANT?

> Architects agree the design with stakeholders before building; diagrams make the design reviewable.

WHERE IS IT USED IN INDUSTRY?

Architects and FDEs present these diagrams to customers.

HOW DOES IT WORK?

- Draw sources on the left and consumers on the right.
- Label every pipeline with its frequency and owner.
- Mark where data is classified and where quality checks run.

PRACTICAL EXAMPLE

> A retail architecture: POS and web data → Kafka → lakehouse bronze/silver/gold → dashboards and an ML feature store.

TOOL OR SOFTWARE

draw.io, Excalidraw, Mermaid

HANDS-ON EXERCISE

Draw the data architecture for a clinic's appointments and billing, with one quality check at each layer.

COMMON MISTAKES

Drawing tools without showing data ownership and security.

INTERVIEW QUESTION

How would you present a data architecture to a CTO?

OFFICIAL AND FURTHER READING

- C4 model for diagrams (official site)

A12.2

System design fundamentals for data and AI systems

WHAT IS IT?

> System design picks components and trade-offs for scale, latency, reliability and cost. The usual steps: clarify requirements, estimate load, design the flow, choose storage, plan failure handling.

WHY IS IT IMPORTANT?

> Interviews for data engineers and FDE roles use system design questions heavily.

WHERE IS IT USED IN INDUSTRY?

Common in technical rounds at data platform and AI companies.

HOW DOES IT WORK?

- Clarify users, data volume, freshness and latency.
- Estimate rows per day and storage per year.
- Choose batch or streaming; choose storage by access pattern.
- Plan retries, monitoring and backfills.

PRACTICAL EXAMPLE

> Design a daily pipeline for 50 million events: batch versus streaming, storage format, partitioning by date, retries and backfills.

TOOL OR SOFTWARE

Whiteboard or document

HANDS-ON EXERCISE

Write a one-page design for a real-time fraud feature pipeline with trade-offs.

COMMON MISTAKES

Jumping into tools before agreeing requirements.

INTERVIEW QUESTION

How do you design a pipeline that must handle reprocessing of last year's data?

OFFICIAL AND FURTHER READING

- System Design Primer (open source)

Added: Governance, privacy and business consulting

What is added: Governance and privacy (GDPR, India's DPDP Act), responsible AI, business problem-solving, requirements and communication.

B3 Business Problem-Solving and Analytics Consulting (course Module 16 gap)

B3.1

Business KPIs, models and process mapping

WHAT IS IT?

> A KPI is a measured result tied to a goal. A business model explains how money is made; process mapping shows each step of a workflow.

WHY IS IT IMPORTANT?

> Analysts who know the business choose the right metric and find the real problem.

WHERE IS IT USED IN INDUSTRY?

Consultants and analysts in banking, retail and healthcare start here.

HOW DOES IT WORK?

- Start from the goal, then the KPI, then the driver metrics.
- Map the process step by step with owners and times.
- Mark the bottleneck and the cost of delay.

PRACTICAL EXAMPLE

> Loan approval takes 9 days; mapping shows 5 days in document collection. The KPI is time to approval; the driver is documents received on day 1.

TOOL OR SOFTWARE

Miro or draw.io for process maps

HANDS-ON EXERCISE

Map one process you know, with three KPIs and one bottleneck.

COMMON MISTAKES

Choosing vanity metrics.

INTERVIEW QUESTION

What is the difference between a KPI and a driver metric?

OFFICIAL AND FURTHER READING

- Atlassian: process mapping (official guide)

B3.2

Problem discovery: stakeholder interviews, Five Whys and fishbone analysis

WHAT IS IT?

> Stakeholder interviews collect pain points. Five Whys asks why repeatedly to reach a root cause. A fishbone diagram groups causes into categories (people, process, tools, data).

WHY IS IT IMPORTANT?

> Solving the symptom wastes money; the root cause is what to fix.

WHERE IS IT USED IN INDUSTRY?

Used in operations, quality and consulting projects.

HOW DOES IT WORK?

- Interview 5 to 8 people across roles.
- Write a problem statement: who, what, impact, how often.
- Five Whys: ask why until the cause is a process or system you can change.
- Fishbone: list causes under each category.

PRACTICAL EXAMPLE

> Problem: late deliveries. Why 1: stock is out. Why 2: forecast is wrong. Why 3: sales data is a week late. Root cause: weekly data load.

TOOL OR SOFTWARE

Whiteboard, Miro

HANDS-ON EXERCISE

Run a Five Whys on a problem from your work and write the root cause.

COMMON MISTAKES

Stopping at the first answer.

INTERVIEW QUESTION

How would you find the root cause of rising returns?

OFFICIAL AND FURTHER READING

- ASQ: root cause analysis (official)

B3.3

Hypotheses, analytical questions and recommendations

WHAT IS IT?

A hypothesis is a testable statement. Analysis answers a question with data; a recommendation says what to do and what result to expect.

WHY IS IT IMPORTANT?

Recommendations that are not tied to a question and a measure are ignored.

WHERE IS IT USED IN INDUSTRY?

Consultants and analysts present recommendations to executives.

HOW DOES IT WORK?

- Write the question, then the hypothesis, then the data needed.
- Test it with the data; state what would prove it wrong.
- Recommend an action with an owner and a measure.

PRACTICAL EXAMPLE

Hypothesis: a reminder email reduces missed appointments by 15%. Test with an A/B split over four weeks.

TOOL OR SOFTWARE

Excel, SQL, Python

HANDS-ON EXERCISE

Write three hypotheses for a shop's falling sales and the data test for each.

COMMON MISTAKES

Recommendations without expected impact.

INTERVIEW QUESTION

How do you turn an analysis into a recommendation?

OFFICIAL AND FURTHER READING

- Harvard Business Review: data-driven decisions (learning article)

B4 Requirements, Communication and Delivery

B4.1

Requirements, user stories, acceptance criteria and PRDs

WHAT IS IT?

> Requirements say what the solution must do. User stories describe value from the user's view. Acceptance criteria say how to test it. A PRD (product requirements document) gathers these.

WHY IS IT IMPORTANT?

> Clear requirements stop rework and disputes with stakeholders.

WHERE IS IT USED IN INDUSTRY?

Product, analytics and BI teams write these every week.

HOW DOES IT WORK?

- Story: As a [role], I want [need], so that [benefit].
- Acceptance: Given [context], when [action], then [result].
- Non-functional requirements: speed, security, accuracy, availability.

PRACTICAL EXAMPLE

> As an operations manager, I want a daily late-order list, so that I can call customers before 10:00. Given an order is late by 2 hours, then it appears on the list by 08:00.

TOOL OR SOFTWARE

Jira, Confluence

HANDS-ON EXERCISE

Write five user stories with acceptance criteria for a dashboard.

COMMON MISTAKES

Stories without testable criteria.

INTERVIEW QUESTION

What is an acceptance criterion?

OFFICIAL AND FURTHER READING

- Atlassian: user stories (official guide)

B4.2

Communicating insights and data storytelling

WHAT IS IT?

> Data storytelling states the main message first, shows the evidence in one chart, and ends with the decision needed.

WHY IS IT IMPORTANT?

> Executives decide from short, clear messages. Long reports go unread.

WHERE IS IT USED IN INDUSTRY?

Analysts, FDEs and consultants present to leaders weekly.

HOW DOES IT WORK?

- Start with the answer.
- One chart per message.
- Name the decision, the owner and the date.

PRACTICAL EXAMPLE

> Slide: Churn rose 4 points in March, driven by two plans. Decision: fix onboarding for those plans by May.

TOOL OR SOFTWARE

PowerPoint, Power BI

HANDS-ON EXERCISE

Turn a 10-slide report into a 3-slide summary with a decision.

COMMON MISTAKES

Showing every chart on the dashboard.

INTERVIEW QUESTION

How would you present bad news to a manager?

OFFICIAL AND FURTHER READING

- Storytelling with Data (official site)

B4.3

Agile and project delivery for analytics teams

WHAT IS IT?

> Agile delivers in short cycles with a backlog, sprints and reviews. Kanban shows flow on a board. Delivery management tracks scope, risk and dependencies.

WHY IS IT IMPORTANT?

> Analytics work is often unclear at the start; short cycles show progress and adjust scope.

WHERE IS IT USED IN INDUSTRY?

Used by data and product teams across industries.

HOW DOES IT WORK?

- Backlog ordered by value.
- Sprint of 1 to 2 weeks with a goal.
- Review with stakeholders; retrospect and improve.

PRACTICAL EXAMPLE

> A two-week sprint delivers the first version of a sales dashboard; the review finds two new metrics to add.

TOOL OR SOFTWARE

Jira, Trello

HANDS-ON EXERCISE

Plan a four-week analytics project in four sprints with goals.

COMMON MISTAKES

Planning without stakeholder reviews.

INTERVIEW QUESTION

How do you run an analytics project in sprints?

OFFICIAL AND FURTHER READING

- Scrum Guide (official)

B5 Data Governance, Privacy and Responsible AI (course Module 15 gap)

B5.1

Data ownership, stewardship and retention

WHAT IS IT?

> An owner decides who may use a dataset; a steward keeps it accurate. Retention sets how long data is kept.

WHY IS IT IMPORTANT?

> Clear ownership makes fixes happen; retention limits risk and cost.

WHERE IS IT USED IN INDUSTRY?

Required in banking, insurance and health data.

HOW DOES IT WORK?

- Each dataset has one owner and one steward.
- Retention period written in policy.
- Deletion runs on schedule.

PRACTICAL EXAMPLE

> Customer call recordings kept 90 days, then deleted by a scheduled job with a log.

TOOL OR SOFTWARE

Catalogue tools (see A4.3)

HANDS-ON EXERCISE

Write a data ownership sheet for five tables in a sales system.

COMMON MISTAKES

No owner for shared tables.

INTERVIEW QUESTION

Who should own customer data in a bank?

OFFICIAL AND FURTHER READING

- DAMA DMBOK overview (official body)

B5.2

Privacy: GDPR and India's Digital Personal Data Protection Act

WHAT IS IT?

> Privacy laws set rules for collecting, using, storing and deleting personal data. GDPR applies to EU residents' data; India's DPDP Act 2023 covers digital personal data in India.

WHY IS IT IMPORTANT?

> Breaches and fines follow poor privacy design. Analysts must minimise and protect personal data.

WHERE IS IT USED IN INDUSTRY?

Every customer-data project in India or Europe.

HOW DOES IT WORK?

- Collect only what is needed (data minimisation).
- Record consent and purpose.
- Give people access, correction and deletion rights where the law requires.
- Report breaches as the law requires.

PRACTICAL EXAMPLE

> A marketing dataset removes phone numbers not needed for the campaign and keeps a consent flag for each contact.

TOOL OR SOFTWARE

Data inventory sheet

HANDS-ON EXERCISE

Review a sample dataset and list the personal fields, purpose and retention for each.

COMMON MISTAKES

Copying production personal data into test files.

INTERVIEW QUESTION

What is data minimisation?

OFFICIAL AND FURTHER READING

- GDPR text (official EU site)
- MeitY: Digital Personal Data Protection Act (official)

B5.3

Data masking, row-level security and access control

WHAT IS IT?

> Masking hides sensitive values (for example showing the last four digits). Row-level security limits which rows a user sees.

WHY IS IT IMPORTANT?

> Analysts need data without exposing personal details.

WHERE IS IT USED IN INDUSTRY?

Used in banks, HR and healthcare analytics.

HOW DOES IT WORK?

- Mask in queries or in the database.
- Power BI and database RLS filter rows by user.
- Audit who accessed sensitive tables.

PRACTICAL EXAMPLE

> A regional manager sees only their region's rows in the Power BI report through RLS.

TOOL OR SOFTWARE

Power BI RLS, PostgreSQL row-level security

HANDS-ON EXERCISE

Set up a row filter for two regions and test it with two users.

COMMON MISTAKES

Relying on report-level hiding instead of security at the data layer.

INTERVIEW QUESTION

How do you stop an analyst from seeing salary data they should not see?

OFFICIAL AND FURTHER READING

- Power BI row-level security (official)

B5.4

Responsible AI, threat modelling and incident response

WHAT IS IT?

> Responsible AI covers fairness, transparency, human oversight and security. Threat modelling lists assets, threats and controls. Incident response plans how to act when something goes wrong.

WHY IS IT IMPORTANT?

> AI systems can leak data, be manipulated (prompt injection) or treat groups unfairly.

WHERE IS IT USED IN INDUSTRY?

Required for AI products in regulated sectors.

HOW DOES IT WORK?

- Check model performance by group.
- Threat model: data, prompts, tools, outputs.
- Plan detection, containment, fix and communication for incidents.

PRACTICAL EXAMPLE

> A chatbot leaks an internal price list through prompt injection; the incident plan disables the tool, rotates keys and updates guardrails.

TOOL OR SOFTWARE

OWASP Top 10 for LLM Applications

HANDS-ON EXERCISE

Write a threat model for an AI chatbot with five threats and controls.

COMMON MISTAKES

Treating guardrails as a complete solution.

INTERVIEW QUESTION

How would you test an AI assistant for prompt injection?

OFFICIAL AND FURTHER READING

- OWASP Top 10 for LLM Applications (official)
- NIST AI Risk Management Framework (official)

Added: Linux, documentation, troubleshooting and roadmap

What is added: Command line, technical documentation, production troubleshooting and the 12-month roadmap.

B9 Linux and the Command Line (Part C gap)

B9.1

Linux and terminal essentials for data work

WHAT IS IT?

> Linux is the server operating system most data platforms run on. The command line lets you manage files, processes and services.

WHY IS IT IMPORTANT?

> Cloud servers, containers and CI runners are Linux; you need the terminal to debug them.

WHERE IS IT USED IN INDUSTRY?

Data engineering and DevOps roles.

HOW DOES IT WORK?

- Navigate: pwd, ls, cd, find, grep.
- Files: cat, head, tail -f, cp, mv, chmod.
- Processes: ps, top, kill.
- Pipes and redirection: |, >, >>.

PRACTICAL EXAMPLE

> tail -f pipeline.log | grep ERROR shows errors live as the pipeline runs.

TOOL OR SOFTWARE

Any Linux or WSL terminal

HANDS-ON EXERCISE

Find all ERROR lines in a log file from the last 1,000 lines and count them.

COMMON MISTAKES

Running commands from unknown sources.

INTERVIEW QUESTION

How would you find which process is using port 8080?

OFFICIAL AND FURTHER READING

- The Missing Semester (MIT, free)
- Linux command reference (free book)

B10 Technical Documentation and Production Troubleshooting (Part C gap)

B10.1

Technical documentation and runbooks

WHAT IS IT?

> Documentation explains what a system does, how to run it and how to fix it. A runbook is a step-by-step guide for common incidents.

WHY IS IT IMPORTANT?

> Handover and on-call work fail without documentation.

WHERE IS IT USED IN INDUSTRY?

Required in every production team.

HOW DOES IT WORK?

- README: purpose, setup, run, owners.
- Data dictionary: tables, columns, meanings.
- Runbook: symptom, check, fix, escalation.

PRACTICAL EXAMPLE

> A runbook for 'dashboard shows yesterday's numbers': check the pipeline run, re-run the failed task, notify the owner.

TOOL OR SOFTWARE

Markdown, Confluence

HANDS-ON EXERCISE

Write a runbook for one failure in your project.

COMMON MISTAKES

Documentation written once and never updated.

INTERVIEW QUESTION

What should a runbook contain?

OFFICIAL AND FURTHER READING

- Google SRE book: effective troubleshooting (free)

B10.2

Production troubleshooting for data pipelines and AI apps

WHAT IS IT?

> Troubleshooting finds why a live system fails: check what changed, read logs, reproduce, fix, verify and review.

WHY IS IT IMPORTANT?

> Stakeholders judge data teams on how fast and calmly they recover.

WHERE IS IT USED IN INDUSTRY?

On-call data engineers and FDEs.

HOW DOES IT WORK?

- Confirm impact and mitigate first (rerun, roll back, switch source).
- Check recent changes, logs and upstream freshness.
- Fix the root cause and add a test or alert.
- Write a short post-incident review.

PRACTICAL EXAMPLE

> A dashboard is wrong: an upstream schema change renamed a column; the fix maps the new name and a schema-change alert is added.

TOOL OR SOFTWARE

Airflow logs, dbt docs, cloud logs

HANDS-ON EXERCISE

Inject a broken schema change in a sample pipeline and write the incident note.

COMMON MISTAKES

Fixing without finding the cause.

INTERVIEW QUESTION

A pipeline failed at 3 am. What do you do first?

OFFICIAL AND FURTHER READING

- PagerDuty incident response guide (official)

C8 Realistic 12-Month Roadmap (beginner)

Item	Answer
Months 1 to 2: Foundation	Excel (module 1), SQL basics (module 2), basic statistics (module 7 start), business analytics basics. About 10 hours a week. Milestone: a cleaned dataset with a 5-metric summary. Assessment: a timed SQL test and an Excel cleaning exercise.
Months 3 to 4: Business intelligence	Power BI (module 3), star schema and DAX, dashboard design. About 10 hours a week. Milestone: a published dashboard with row-level security. Assessment: dashboard review against a rubric.
Months 5 to 6: Programming and data	Python (module 6), pandas, visualisation, Git and GitHub. About 12 hours a week. Milestone: a Python EDA on a public dataset in GitHub. Assessment: code review with tests.

Months 7 to 8: Machine learning and statistics	Statistics (module 7), supervised learning, evaluation, a churn model. About 12 hours a week. Milestone: a churn model with a business recommendation. Assessment: model evaluation and explanation.
Months 9 to 10: Data engineering and cloud	Warehouse concepts, ETL and ELT, dbt, Airflow, one cloud (see Supplement A). About 12 hours a week. Milestone: a three-layer pipeline with tests. Assessment: pipeline review and a failure drill.
Months 11: GenAI, agents and MCP	LLM APIs, RAG, agents, MCP (modules 5 and 8). About 12 hours a week. Milestone: a RAG assistant with citations and an evaluation. Assessment: evaluation report.
Month 12: Portfolio and interviews	Two capstone projects published, resume, LinkedIn, mock interviews and the readiness check (C9). About 15 hours a week. Milestone: three mock interviews and two published projects. Assessment: interview rubric.
What employers generally expect	Working projects with code and data, clear documentation, measured results, the ability to explain trade-offs and a steady learning record. The programme makes no promise of employment or any salary.